Course Name: Operating System		**Midterm Exam**	
Course Code : CS201		Date :	19/11/ 2025
Level : 2		Time :	1 Hour
Program : Artificial Intelligence and Data Science	**Zagazig National University**	No. of Page :	2 pages
Term No. : first	**The Engineering Sector**	No. of Questions :	22
		Full Mark :	15

• الاسم:.....

Choose the correct answer:-

1. **In the context of operating systems, what is the main purpose of a kernel?**

A. To provide a user interface for interacting with the system	B. To manage the system's CPU, memory, and other hardware resources
C. To compile and execute user programs	D. To create and manage network connections

2. **In a time-sharing operating system, CPU scheduling is important to:**

A. Minimize memory usage by each process	B. Allow multiple users to interact with the computer simultaneously
C. Restrict the number of active processes to one	D. Reduce the need for peripheral devices

3. **What is the main purpose of a mode bit in an operating system?**

A. To indicate whether the CPU is in user or kernel mode	B. To improve memory access speed
C. To manage file permissions	D. To signal the end of a process

4. **The purpose of protection in an operating system is to:**

A. Control access to system resources by processes	B. Restrict the types of applications that can be installed
C. Ensure efficient resource allocation	D. Manage network communication between systems

5. **System programs that act as intermediaries between users and hardware components are known as:**

A. Interpreters	B. Device drivers
C. Compilers	D. API libraries

6. **The primary disadvantage of a microkernel system structure is:**

A. Increased risk of system crashes	B. Larger codebase leading to memory bloat
C. Decreased efficiency due to frequent communication between components	D. Lack of support for modularity

7. **Which system program allows users to view and edit files in an operating system?**

A. Device manager	B. Text editor
C. File system manager	D. API

8. **What is the primary purpose of the API (Application Programming Interface) in the context of operating systems?**

A. To provide direct access to hardware	B. To give user programs the ability to request OS services in a standardized way
C. To manage low-level kernel tasks	D. To optimize CPU scheduling

9. **In the context of an operating system, which of the following best describes a "trap"?**

A. A software-generated interrupt that is handled by the operating system	B. A hardware error that forces the system to shut down
C. A method of switching from kernel mode to user mode	D. A network error that prevents resource sharing

10. **What is the purpose of the kernel mode and user mode in an operating system?**

A. To distinguish between system and application programs	B. To protect critical system resources from unauthorized access by applications
C. To manage multi-threading	D. To optimize memory usage for the OS

11. **The function of the wait system call in process management is to:**

A. Pause the process until resources are available	B. Allow a process to sleep for a specified time
C. Suspend the execution of a parent process until a child process has terminated	D. Swap a process out of memory

12. **Which of the following system calls is used to create a new process in most operating systems?**

A. Open	B. Exec
C. Fork	D. read

13. **Which of the following is the first step in process creation?**

A. Allocation of memory	B. Loading program into memory
C. Assigning a unique process ID	D. Creating a process control block (PCB)

14. **Which state does a process enter after being created and before it starts executing?**

A. Ready	B. New
C. Running	D. Blocked

15. **Which system call is used to terminate a process in UNIX?**

A. exit()	B. fork()
C. wait()	D. kill()

16. **Which of the following is the primary function of the long-term scheduler?**

A. To decide which process to execute next.	B. To load processes into memory.
C. To select processes from the job pool.	D. To allocate CPU time to the processes.

17. **What is the role of the short-term scheduler in an operating system?**

A. It selects processes for execution from the ready queue	B. It creates new processes.
C. It loads the program into memory.	D. It manages long-term process scheduling

18. **Which of the following is an example of a long-term scheduler's action?**

A. Deciding which process to load from the disk into memory.	B. Allocating CPU time to a process.
C. Switching between running processes.	D. Managing process queues.

1) List the different states of a process in an operating system.

- **Answer**: The different states of a process are:
  - **New**: Process is being created.
  - **Ready**: Process is waiting for CPU time.
  - **Running**: Process is currently executing on the CPU.
  - **Waiting (Blocked)**: Process is waiting for some event to occur (e.g., I/O completion).
  - **Terminated**: Process has finished execution.

2) List the scheduling queues of processes

- **Job Queue**
- **Ready Queue**
- **Device Queue**

3) what are the main types of pipes

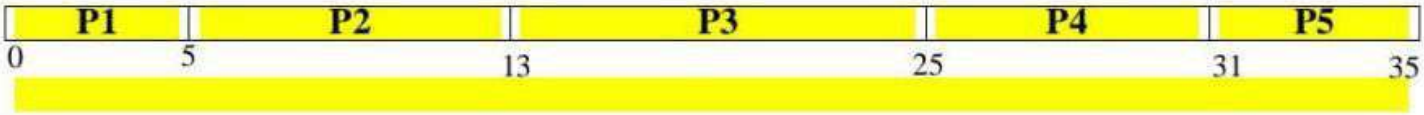
1. **ordinary Pipes**(anonymous pipes)
2. **Named Pipes**

4) Suppose we have five processes with the following **burst times**:

Process	Burst Time
P1	5
P2	8
P3	12
P4	6
P5	4

Suppose that the processes arrive in the order: P1 , P2 , P3 ,P4,P5

A. **what is The Gantt Chart for the schedule ?**
B. **What is the waiting time for each process?**
C. **What is the Average waiting time?**



the waiting time for each process is

P1=0,P2=5,P3=13,P4=25,P5=31

the Average waiting time is

(0+5+13+25+31)/5=14.8